

SPORTS INJURY RISK ASSESSMENT REPORT

AI-Powered Movement Biomechanics & Predictive Injury Intelligence (Milestone 3)

OVERALL INJURY RISK: MODERATE
(32.5%)

Model: Trained Random Forest & XGBoost
(ROC-AUC: 0.807)

1. Athlete Demographics & Video Session Details

Athlete Name:	Gopal M V	Sport / Position:	Football • Forward / Striker
Age / Gender:	22 yrs	Height / Weight:	180 cm / 75 kg
Exercise Activity:	SQUAT	Pose Detection Rate:	98.5% (180 frames)
Injury History:	Prior left ankle sprain	Assessment Date:	31 Aug 2026, 07:06 AM

2. Machine Learning Injury Category Forecast (Module 6)

Injury Category	Predicted Risk %	Primary Risk Trigger	Severity Tier
ACL Tear Risk	36%	Knee valgus angle & bilateral landing asymmetry	Normal
Hamstring Strain Risk	29%	Hip extension velocity & rapid deceleration torque	Normal
Ankle Sprain Risk	34%	Lateral foot inversion & landing instability	Normal
Lower Back Pain Risk	28%	Excessive forward trunk lean & spinal shear force	Normal
Shoulder Dislocation Risk	21%	Arm swing elevation & upper torso rotation	Normal

3. Biomechanical Joint Kinematics Breakdown (MediaPipe 3D)

Joint Structure	Min Flexion	Max Extension	Range (ROM)	Kinetic Status
Left Knee Joint	82.5°	172.0°	89.5°	Symmetrical
Right Knee Joint	94.0°	171.5°	77.5°	Symmetrical
Left Hip Joint	78.0°	165.0°	87.0°	Symmetrical
Right Hip Joint	81.0°	164.0°	83.0°	Symmetrical
Left Elbow Joint	Not available	Not available	Not available	Symmetrical
Right Elbow Joint	Not available	Not available	Not available	Symmetrical

Knee Symmetry: 86.6% | Hip Symmetry: 95.4% | Mean Trunk Lean: 14.5° | Consistency: 91.2%

4. AI-Prescribed Corrective Rehabilitation & Training Plan (Module 9)

- AI-Prescribed Rehabilitation: Strength Training & Single-Leg Loading (Est. Recovery: 4 weeks)
- Incorporate single-leg Romanian deadlifts to balance 13.4% knee ROM asymmetry.
- Core plank progressions to stabilize spinal lean under fatigue.

Certification: Generated by the Sports Injury Risk Detection & Prevention Platform (Infosys Springboard Milestone 3). Evaluated with Google MediaPipe 33-Keypoint Pose Tracking and Supervised Machine Learning (Random Forest & XGBoost). Designed for athletic screening and performance optimization; consult a certified physiotherapist for medical diagnosis.